

1. Consider the integral $\int_0^{\pi} x \cos(x) dx$. Which of the following expressions represents this integral as a limit of Riemann sums using right endpoints? Circle the correct answer.

A). $\lim_{n \rightarrow \infty} \sum_{i=1}^n \left[\frac{\pi i}{n} \cos\left(\frac{\pi i}{n}\right) \cdot \frac{\pi i}{n} \right]$

B). $\lim_{n \rightarrow \infty} \sum_{i=1}^n \left[\frac{\pi}{n} \cos\left(\frac{\pi i}{n}\right) \cdot \frac{\pi}{n} \right]$

C). $\lim_{n \rightarrow \infty} \sum_{i=1}^n \left[\frac{\pi i}{n} \cos\left(\frac{\pi i}{n}\right) \cdot \frac{\pi}{n} \right]$

D). $\lim_{n \rightarrow \infty} \sum_{i=1}^n \left[\frac{\pi}{n} \cos\left(\frac{\pi}{n}\right) \cdot \frac{\pi}{n} \right]$

2. Consider the function $f(x) = 12 - x^2 + x$ on the interval $[-3, 3]$. The graph of the function is given below.

- 1). Sketch the appropriate rectangles on the graph to illustrate the right-endpoint Riemann sum R_3 for $f(x)$ on the interval $[-3, 3]$ with $n = 3$.
- 2). Calculate the right-endpoint Riemann sum R_3 for $f(x)$ on the interval $[-3, 3]$ with $n = 3$.

